Lesson 3.4 Guided Notes

Session 1 · Graphing Ratios on the Coordinate Plane

I can turn a ratio into ordered pairs and plot them on a coordinate plane. · 6.RP.A.3a

Word / Spanish	Meaning — complete the blank	Example
coordinate plane / plano de coordenadas	A _____ where you plot points.	x across, y up
ordered pair / par ordenado	(x, y) names one point: _____ first, then up.	(2, 14)
origin / origen	The point _____ where the axes cross.	0 trays, 0 muffins

Key idea

Each column of a ratio table becomes one _____: x from the top row, y from the bottom row.

Points from equivalent ratios lie on a _____ line through the _____.

Worked example

Each tray holds 8 muffins. Let x = trays and y = muffins. Graph the ratio.

Table: 1 tray = 8 muffins, 2 trays = _____, 3 trays = _____.

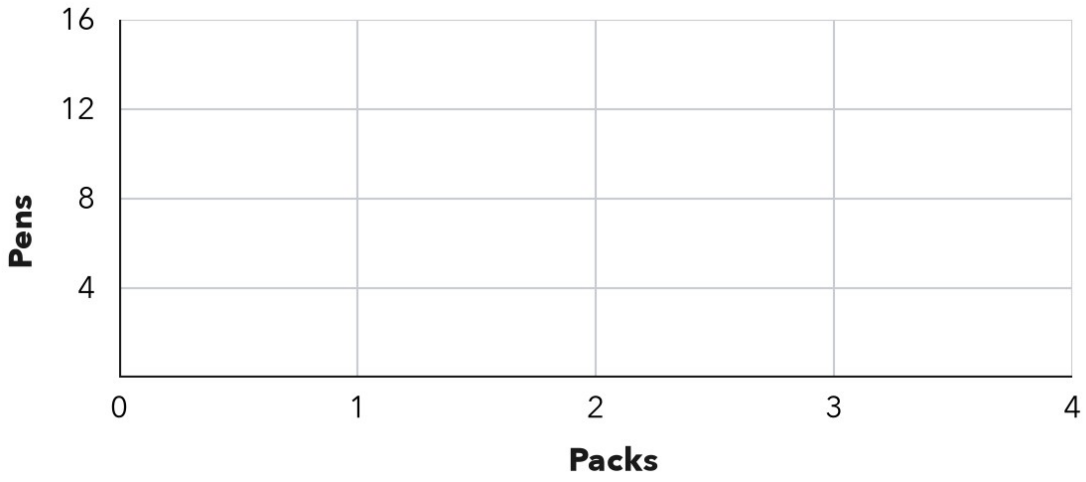
Pairs: (1, 8), (2, 16), (3, 24). Go across x, then up y.

Plot each pair, starting from (0, 0).

Answer: The points lie on a _____ through (0, 0).

You try

1. Each pack holds 4 pens (x = packs, y = pens). Write the pairs for 1, 2 and 3 packs. Plot them.



_____

2. The ratio of x to y is 1 : 7. Write the ordered pair for:
a. x = 2 _____
b. x = 5 _____

_____

Sum it up: On a ratio graph, the point (x, y) means ___, and the line always passes through ___.

_____

Session 2 · Interpreting Points on Ratio Graphs

I can explain what a point on a ratio graph means in the situation it describes. · 6.RP.A.3a

Word / Spanish	Meaning — complete the blank	Example
unit rate point / punto de tasa unitaria	The point (1, r). The r is the amount for _____.	(1, 9): 9 for each 1
origin / origen	(0, 0): zero of one quantity goes with _____ of the other.	0 hours, 0 miles
interpret / interpretar	To say in _____ what a point means.	(4, 36): 4 driveways earn $36

Key idea

A point (x, y) means x of the first quantity goes with _____ of the second. Read the axis labels.

To find the unit rate point (1, r), divide _____ by x.

Worked example

Let x = pounds of apples and y = cost in dollars. The line passes through (4, 12). How much do 6 pounds cost?

Read the point: 4 pounds cost _____.

Divide: 12 ÷ 4 = _____ dollars per pound, so the unit rate point is (1, 3).

Multiply: 6 × 3 = 18.

Answer: 6 pounds cost _____; the point is (6, 18).

You try

1. Find the unit rate point (1, r) for a line through:
a. (5, 35) _____
b. (2, 9) _____

_____

2. A graph of hours (x) and dollars earned (y) passes through (3, 45). What does (3, 45) mean? What is the pay for 1 hour?

_____

Sum it up: The point (1, r) on a ratio graph tells me ___ because ___.

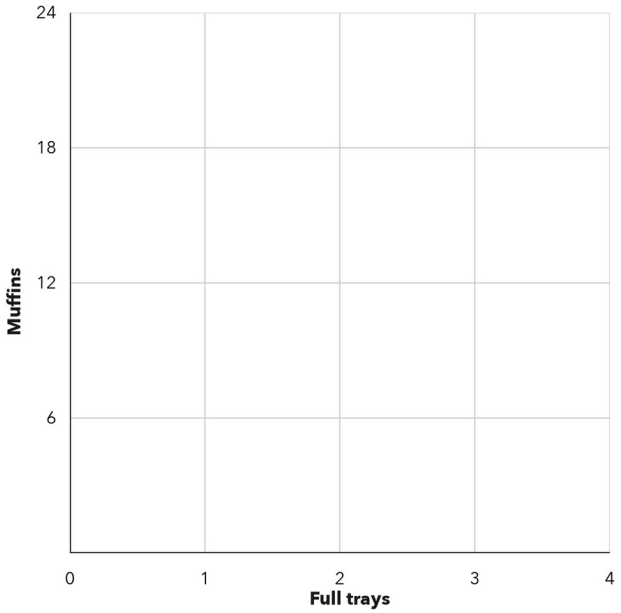
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Session 1 · Model, reason, and explain

Graphing Ratios on the Coordinate Plane · 6.RP.A.3a

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Each full tray holds 6 muffins. Complete a table for 0, 1, 2, and 3 full trays, then plot its points. Label both axes and explain the origin.



_____

_____

_____

Extend your reasoning

Is (1.5,9) a possible order if only full trays are sold?

_____

_____

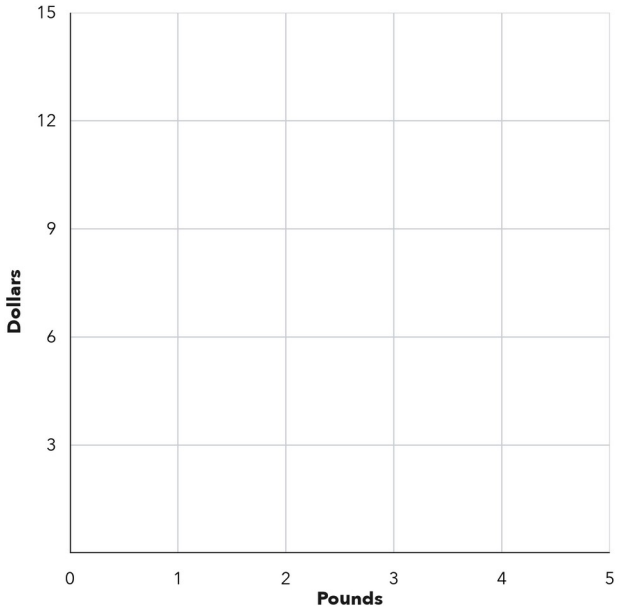
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Interpreting Points on Ratio Graphs · 6.RP.A.3a

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Apples cost $3 per pound. Plot (1,3),(2,6),(4,12). Explain (4,12), decide whether (3,10) belongs to the relationship, and find the cost of 2.5 pounds.



_____

_____

_____

Extend your reasoning

Compare why intermediate points are meaningful for apples but not full trays of muffins.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?